

A Novel 1T-APS Using 22 nm FD-SOI Technology With Record High Tunable Sensitivity Range

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Abstract—A novel one transistor active pixel sensor (1T-APS) with extraordinary tunable sensitivity range is experimentally demonstrated using 22 nm fully depleted silicon-on-insulator (FD-SOI) technology. The proposed device enables independent control of both full well capacity and sensitivity. Two electrodes are connected to the substrate, with one used to form the depletion region beneath the buried oxide (BOX) and the other to tune the movement of photoelectrons. The experimental results show a record high tunable sensitivity range of 340000%, varying from 4 mV/($\mu\text{J}/\text{cm}^2$) to 13.6 V/($\mu\text{J}/\text{cm}^2$). With tunable sensitivity, the dynamic range (DR) is expanded from 51.5 dB to 101.5 dB, making it promising for high-dynamic-range imaging applications.

Index Terms—1T-APS, tunable sensitivity, high dynamic range, FD-SOI, 22 nm technology.

I. INTRODUCTION

PHOTO detectors, represented by charge coupled device (CCD) and complementary metal-oxide-semiconductor (CMOS) image sensor (CIS), are widely used in modern electronics [1], [2], [3], [4]. The traditional 4T-APS based CIS utilizing four transistors for processing, amplification, and gating, stands out in competition with CCD for its lower cost and power consumption but suffers from complicated

Received 26 March 2025; revised 21 April 2025; accepted 22 April 2025. Date of publication 24 April 2025; date of current version 1 July 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA1200500, in part by the National Natural Science Foundation of China under Grant 62474052 and Grant 62404055, in part by the Natural Science Foundation of Shanghai under Grant 23ZR1405900, in part by Guangdong Province Research and Development in Key Fields from Guangdong Greater Bay Area Institute of Integrated Circuit and System under Grant 2021B0101280002, in part by Guangzhou City Research and Development Program in Key Field under Grant 20210302001, and in part by Guangdong Key Laboratory of Integrated Circuit Technology and Products Based on Fully Depleted Silicon on Insulator. The review of this letter was arranged by Editor R.-H. Horng. (Corresponding authors: Yu-Long Jiang; Yong Xu; Jing Wan.)

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Digital Object Identifier 10.1109/LED.2025.3563988

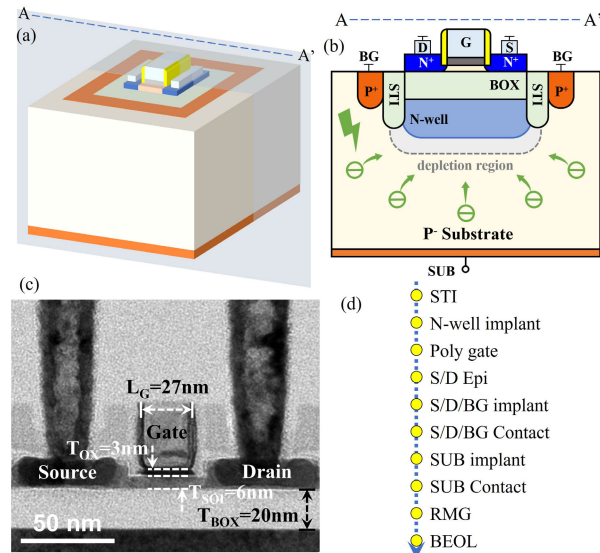


Fig. 1. (a) Schematic 3D structure and (b) 2D cross-section of TPISD. (c) XTEM image and (d) the simplified process flow.

pixel structure and low fill factor (FF) [5], [6]. The 1T-APS greatly improves the FF by integrating all these functions into single transistor with high pixel density [3], [7], [8], [9], [10], [11], [12], [13]. As the digital cameras keep pursuing higher pixel density, developing pixels with higher performance and more functions becomes very attractive, especially for high-dynamic-range imaging [14], [15], [16], [17], [18], [19].

One effective approach is to make sensitivity tunable. For instance, higher sensitivity is often required for dark environments to detect the weak light, while lower sensitivity is preferable for brighter environment to avoid overexposure, considering the constant full well capacity (FWC) [11], [20], [21]. Previous reports utilized additional structures, such as photogate/photodiode [20], double-photodiode feedback [22], gate/body-tied photodetector and transfer gate [23], [24], etc., to regulate the sensitivity. However, these increase the fabrication complexity and reduce the FF. Photoelectron in-situ sensing device (PISD), a novel 1T-APS based on deep depletion and interface coupling effect, has been experimentally proven to exhibit high sensitivity [10] and achieve a tunable sensitivity range of 40% by introducing two additional control gates [11].

In this letter, a new 1T-APS named tunable PISD (TPISD) is demonstrated, by integrating an extra bottom electrode. The independent regulation of FWC and sensitivity is realized, and the record-high tunable sensitivity range of 340000% is experimentally demonstrated.

II. DEVICE FABRICATION

Figs. 1 (a) and (b) schematically depict the 3D structure and the 2D cross-section of TPISD. The active region is surrounded by a back gate (BG) electrode placed on a p^+ -type region while a substrate (SUB) electrode is formed at the bottom. An n-well is formed beneath the BOX to act as the embedded photodiode with p-substrate. V_{BOX} , which determines the FWC, is solely controlled by V_{BG} since it is short-circuited with BG electrode, while V_{SUB} generates an electric field to regulate the photoelectrons.

By adjusting V_{BG} and V_{SUB} , photoelectrons can be extracted from or pushed into the depletion region, resulting in the varying photo responses. The cross-sectional transmission electron microscopy (XTEM) image and process flow of the fabricated device are shown in Figs. 1 (c) and (d). The fabricated device features a gate length of 27 nm, with 6 nm top Si, 3 nm gate oxide (HfO_2) and 20 nm BOX. The total thickness is 775 μm .

The process begins with a hybrid SOI substrate formed by etching and epitaxy, followed by shallow trench isolation (300 nm STI) to define the active region. An n-well beneath the BOX is formed through ion implantation, with deposition and etching used to define the poly gate. Next, epitaxy is employed to form the raised source and drain, while ion implantation and metal deposition are utilized to form electrode contacts. The SUB electrode is formed with backside ion implantation and metal deposition. Finally, the poly gate is replaced with metal gate using the replacement metal gate (RMG) process.

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

Fig. 2 (a) illustrates the current-voltage (IV) characteristics of I_D vs. V_{BG} and V_{SUB} . The I_D - V_{BG} curve aligns with the typical MOSFET transfer characteristic, while I_D hardly varies with V_{SUB} , indicating that the potential below the BOX is mainly controlled by V_{BG} instead of V_{SUB} . Fig. 2(b) demonstrates the drain current variation of TPISD during photodetection. Initially, at $t=20$ ms, negative V_{BG} and V_{SUB} pulses are applied to create a depletion region beneath the BOX. At $t=40$ ms, light with a wavelength of 520 nm and an intensity of $100 \mu W/cm^2$ is turned on, leading to the accumulation of photoelectrons beneath the BOX and variation in drain current. For a constant $V_{BG} = -5$ V, the slope of I_D significantly increases as $|V_{SUB}|$ increases. When $V_{SUB} = -5$ V, I_D saturates at $t=90$ ms after exposure, whereas the saturation time are 3.2 ms and more than 80 ms for $V_{SUB} = -6$ V and $V_{SUB} = -4$ V, respectively, indicating the regulation of sensitivity.

The threshold voltage (V_{TH}) variation is measured using a source follower. Fig. 2(b) depicts the measurement setup. V_D and V_G are biased at 0.8 V, while the source is connected to a 10 k Ω resistor. The output voltage, V_{OUT} , reflects variation of threshold voltage (V_{TH}) during measurement as $V_{OUT} \approx V_G - V_{TH}$ [10], [11].

Fig. 2(d) presents the corresponding experimental results. The variation of V_{OUT} aligns well with I_D in Fig. 2(b), exhibiting different sensitivities. Two consecutive photodetection events are demonstrated, confirming the successful reset. However, it must be noticed that the FWC (maximum ΔV_{OUT} before and after illumination) remains unchanged with different V_{SUB} , as it is solely controlled by V_{BG} as shown in Fig. 2 (a). For a constant $V_{SUB} = -6$ V, variations of V_{BG} ($V_{BG} = -6$ V, $V_{BG} = -5$ V, blue line, and $V_{SUB} = V_{BG} =$

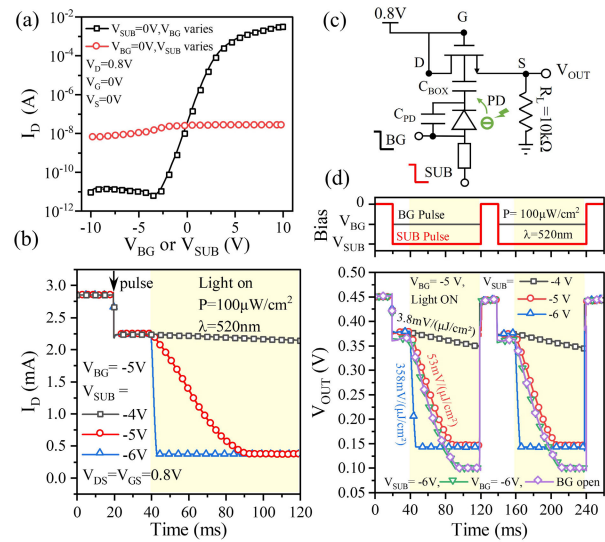


Fig. 2. (a) The DC I_D - V_{BG} and I_D - V_{SUB} characteristics of TPISD. (b) Experimental current variation of TPISD during photo detection. (c) The measurement setup of a source follower formed by TPISD. (d) Experimental photo response showing the individual regulation of FWC and sensitivity regulation with various V_{SUB} .

-6 V, green line) affect both the sensitivity and the FWC, as the former is due to the changes in ΔV_{BG-SUB} , while the latter is attributed to variation in V_{BG} . The response of $V_{SUB} = V_{BG} = -5$ V (red line) and $V_{SUB} = V_{BG} = -6$ V (green line) suggests that for $V_{BG} = V_{SUB}$, only the FWC varies while the sensitivity remains constant. A special case of $V_{SUB} = -6$ V with a floating BG electrode (purple line) was measured too. Its sensitivity shows no difference compared to $V_{SUB} = V_{BG} = -6$ V (green line), indicating that V_{SUB} alone cannot regulate the photoelectrons in the substrate, since the voltage mainly drops at the photo diode junction and the electric field within the substrate is negligible.

Fig. 3(a) shows the variation of V_{OUT} in a dark environment following a $V_{BG} = V_{SUB} = -5$ V pulse. Thermally generated electrons take 225 s to fill the depletion region, whereas only 50 ms is required under light exposure, indicating a high I_{Light}/I_{Dark} ratio of 4500. Fig. 3 (b) illustrates the regulation of FWC, with the 2D color map demonstrating that the FWC is primarily controlled by V_{BG} . The FWC initially increases linearly from 0.09 V at $V_{BG} = -1$ V to 0.32 V at $V_{BG} = -8$ V, and then starts to decrease to 0.28V at $V_{BG} = -10$ V. This decrease occurs because the saturated V_{OUT} gradually approaches 0 V. The 2D color map in Fig. 3 (c) shows a diagonal pattern, indicating that sensitivity is governed by the voltage difference between V_{SUB} and V_{BG} , which determines the electric field in the substrate. For a constant $V_{BG} = -5$ V, the sensitivity is 4 mV/($\mu J/cm^2$) at $V_{SUB} = -1$ V, while it increases to 13.6 V/($\mu J/cm^2$) at $V_{SUB} = -10$ V, this demonstrates an exponential relationship in sensitivity modulation, with a record-high tunable range of 340,000% [11], [19], [20], [22], [23], [24], [25].

Fig. 3 (d) presents the experimental photoelectric responses under various light intensities. The maximum detectable intensity refers to the light intensity that saturates the sensor within a fixed integration time (e.g., 50 ms), while the minimum detectable intensity is the level at which the output signal is not submerged by noise. For the device with a moderate sensitivity ($V_{BG} = V_{SUB}$), the dynamic range is 53.5 dB as shown in

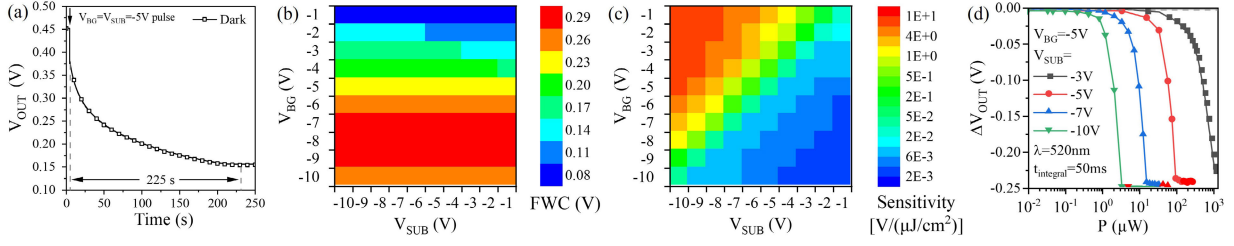


Fig. 3. (a) The V_{OUT} variation in Dark environment after applying $V_{BG} = V_{SUB} = -5V$ pulse. (b) 2D color map showing the relation between FWC and bias voltages (V_{BG} and V_{SUB}). (c) 2D color map showing the relation between sensitivity and bias voltages. (d) Output variation with different light intensities with a fixed integration time of 50 ms.

TABLE I

SUMMARIZES OF THE MAJOR PARAMETERS OF OTHER TYPICAL RELATED WORK

Reference	2006[25]	2008[26]	2016[27]	2018[19]	2019[12, 28]	2019[10]	2020[24]	2021[11]	2025 This work
Pixel Structure	3T	2T	5T+1 Diode	3T+cascode+inverter	1T+ buried diode	1T	5T+floating gate	1T+ 2 control gates	1T+ additional electrode
Transistor Number	3	2	5	8	1	1	5	1	1
Sensitivity	0.53~1.1 V/lux.s	600h+/lux.s	51, 257mV/lux.s	NA	2.33mV/lux.s	1.5 V/(μJ/cm ²)	NA	0.123-0.175 V/(μJ/cm ²)	0.004-13.6 V/(μJ/cm ²)
Tunable range	~100%	0	0	NA	0	0	NA	42.28%	340000%
Dynamic Range(dB)	74.3	52	>106	NA	120-130	NA	NA	NA	>101.5
Function	Sensitivity tunable	-	Linear-log response	Sensitivity tunable	Linear-log response	-	Sensitivity tunable	Sensitivity tunable	Sensitivity tunable
Pixel Pitch	6μm	1.4μm	10 μm	13 μm	2μm	420nm	-	10μm	5.3μm in experiment 1μm in simulation
Process	0.35μm	0.13μm	0.35μm	0.35μm	28nm FDSOI	22nm FDSOI	0.35μm	22nm FDSOI	22nm FDSOI

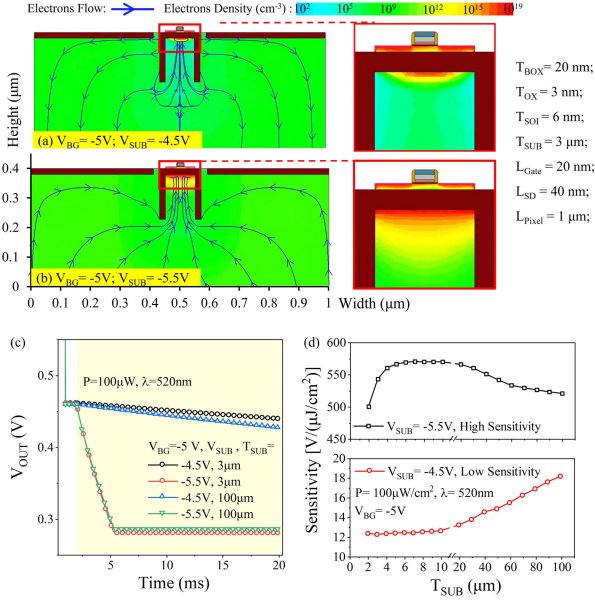


Fig. 4. The electrons flow (left) and electrons distribution beneath the BOX (right) of the simulated device in (a) low sensitivity, (b) high sensitivity modes. The snapshot is captured after 3ms exposure. (c) The simulated tunable photo response under $T_{SUB} = 3\mu m$ and $100\mu m$. (d) The simulated relationship between sensitivity and T_{SUB} .

Eq. (1). With the introduction of the sensitivity regulation, the dynamic range expands to more than 101.5 dB as shown in Eq. (2), showing the significant potential for high dynamic range applications.

$$DR \text{ (no regulation)} = 20 \log \frac{104}{0.22} = 53.5 \text{ dB} \quad (1)$$

$$DR \text{ (with regulation)} > 20 \log \frac{1183}{0.01} = 101.5 \text{ dB} \quad (2)$$

The operation principle and the effect of substrate thickness (T_{SUB}) are further investigated through TCAD simulation, which incorporates field- and doping-dependent mobility, Shockley-Read-Hall thermal generation-recombination with doping-dependent carrier lifetime and photo generation models. The total length of the simulated pixel is $1 \mu m$, including

20 nm gate, 40 nm source and drain. Figs. 4 (a) and (b) illustrate the electrons flow and electrons distribution after 3 ms exposure in TPISD with $3\mu m$ T_{SUB} . In low sensitivity mode ($V_{BG} = -5$ V, $V_{SUB} = -4.5$ V), only a limited number of electrons accumulate under the BOX due to the electric field pushing the photoelectrons away from the device, as shown in Fig. 4 (a). In high sensitivity mode ($V_{BG} = -5$ V, $V_{SUB} = -5.5$ V), the generated electric field drives all photoelectrons in the substrate towards the device, accumulating beneath the BOX and resulting in a high sensitivity, as shown in Fig. 4 (b). The tunable experimental sensitivity with V_{SUB} - V_{BG} is well confirmed by the simulated results shown in Fig. 4 (c). Fig. 4(d) illustrates the simulated sensitivity for different T_{SUB} values. For T_{SUB} smaller than $3\mu m$, sensitivity is influenced by two competing factors. As T_{SUB} increases, photo-absorption increases, generating more carriers and enhancing sensitivity. However, the electric field in the substrate weakens due to $E = V_{BG} - V_{SUB} / T_{SUB}$, reducing the sensitivity modulation window under the same bias conditions. When T_{SUB} exceeds $3 \mu m$, the number of generated carriers remains nearly constant, and further increases in T_{SUB} only reduce the electric field in substrate. However, a high sensitivity modulation window is demonstrated across all substrate thicknesses

Table I compares our TPISD with other related works, highlighting its record-high tunable sensitivity range and simple structure, revealing its promising applications in high dynamic range imaging.

IV. CONCLUSION

A novel 1T-APS with tunable FWC and sensitivity has been experimentally demonstrated. The maximum ΔV_{OUT} , reflecting FWC, is directly controlled by V_{BG} , while sensitivity is tuned by the voltage difference between V_{BG} and V_{SUB} . It is revealed that the FWC can increase from 0.08 V at $V_{BG} = -1$ V to 0.29 V at $V_{BG} = -7$ V, and the sensitivity can be flexibly adjusted from 4 mV/(μJ/cm²) to 13.6 V/(μJ/cm²), achieving a tunable range of 340,000%. The dynamic range is expanded from 53.5 dB to more than 101.5 dB by introducing the tunable sensitivity, making the device promising for high dynamic applications.

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